

Fig S3 | Why the benchmark is built this way (rebuild of the retired null_control against the curated reference only). (a) Against the 18,432,135-entry PolyASite dump with whole-peak-interval matching, the five peak-calling strategies scored 0.996-1.000 (spread < 0.004), while width-matched shuffled peaks already scored 0.62-0.75 (07 1) - and the two retired figures computed different nulls for the same arm (0.615 strand-matched chrom-shuffle vs 0.753 strand-agnostic genome-shuffle; gene-body variant 0.749; gap 7): under that regime the benchmark can rank nothing. Scored at the strand-matched 3'-most base against the curated PolyASite 2.0 representative sites (569,005 clusters, GRCh38.96), the same five arms separate (P@100 0.55-0.492, spread 0.14) over one reconciled null (0.020-0.024, 3 seeds x 5 arms). (b) Reference density recomputed on the curated point reference (dump densities retired): 2.26% of gene-body bp lies within 100 bp of a same-strand curated site, and the measured null mean (0.0219) equals that local density within 3% - chance in this benchmark IS reference density, which is why the null is indispensable. One site per 5.4 kb of genome (568,608 sites; 397 scaffold-only excluded). (c) The same score_tool.py null (width-preserving 3-seed shuffle inside 1.80 Gb of merged gene bodies, scored identically to the real calls) accompanies every verified score TSV: the pre-registered PeakATail default scores 0.7062 / 0.7450 / 0.7572 (PBMC / mouse 1 / mouse 2) against nulls of 0.0217 / 0.0139 / 0.0138 - 33-55x above chance - and every competitor is scored with its own identically-built null. scUTRquant (*) is catalog-based and not ranked; SCAPTURE mouse 2 is not plotted (site-level run scored 0.672 (15 3) but only mouse 1 carries a verified null-bearing row - fig2_accuracy convention). Sources: benchmark_curated.tsv (07 2, SOUND), fig2_accuracy.tsv (19, FIXED), density_recomputed_from_polyasite2.GRCh38.96.rep_sites.bed6; retired-regime constants 05 R4/R5 (FIXED records).